Notes: Bandiera, Barankay & Rasul (QJE, 2007)

Incentives for Managers and Inequality among Workers: Evidence from a Firm-Level Experiment

Contents

1 Big picture

- **Question.** How does tying managers' pay to average lower-tier worker productivity change firm productivity, worker productivity dispersion, and earnings inequality?
- **Experiment.** The authors engineered a mid-season change from fixed managerial wages to performance bonuses based on field-day average worker productivity. Workers' piece-rate contract remained unchanged.
- **Main result.** Mean productivity rises by about 21 percent, while dispersion rises substantially.
- **Mechanisms.** Managers target high-ability workers; the COO selects more able workers into picking and low-ability workers out of picking.
- **Economic takeaway.** Managerial incentives can raise productivity while increasing inequality among observationally similar workers.

2 Incremental contribution

- The paper shifts the incentives literature from worker incentive pay to managerial incentive pay and traces the pass-through to lower-tier workers.
- It opens the black box of the firm by linking one tier's contract to individual outcomes in another tier.
- It shows that performance pay has distributional consequences, not only efficiency consequences.
- It separates targeting and selection mechanisms using worker-level personnel records and daily labor supply data.
- It implements a natural field experiment in a real firm, avoiding endogeneity of observed contracts.

3 Data and measurement

The setting is a UK soft-fruit firm with one COO, field managers, and fruit pickers. Productivity is kilograms picked per hour. The available worker pool is observed daily because seasonal workers live on the farm. The main treatment is a shift in managerial compensation; the worker piece-rate scheme remains fixed.

Interpretation. Observing both productivity and daily availability lets the paper distinguish productivity targeting from selection into work.

4 Model and empirical specifications

Worker output is summarized by

$$y_i = (1 + km_i)e_i, \qquad k > 0,$$

where m_i is managerial effort targeted toward worker i and e_i is worker effort. Managerial pay is

$$w + bY,$$

where Y is average worker productivity. The COO's pay is similarly tied to average productivity.

The main field-day productivity regression is

$$\log Productivity_{ft} = \alpha + \beta Bonus_{ft} + \gamma X_{ft} + \text{field FE} + \text{manager FE} + u_{ft}.$$

The dispersion regression replaces the dependent variable with the log coefficient of variation of productivity.

Interpretation. A positive β in the first equation captures efficiency gains. A positive β in the dispersion equation captures the distributional effect of managerial incentives.

5 Identification and empirical design

- The timing and form of the managerial compensation change were controlled by the researchers.
- Workers' piece-rate contract was unchanged.
- The same workers and managers are observed under both regimes.
- The daily available labor pool is observed, allowing direct study of selection.
- A 2004 placebo season helps address seasonality and crop-cycle concerns.

6 Tables and figures

6.1 Figure I: Time Series of Productivity + Table I: Descriptives of Worker Productivity

Read: Figure I shows the treatment-season productivity increase and the absence of a comparable placebo-season jump. Table I shows mean productivity rising from 8.37 to 10.4 kg/hr and dispersion increasing under the bonus regime.

Economic message. The core fact is simultaneous improvement in mean productivity and widening productivity dispersion.

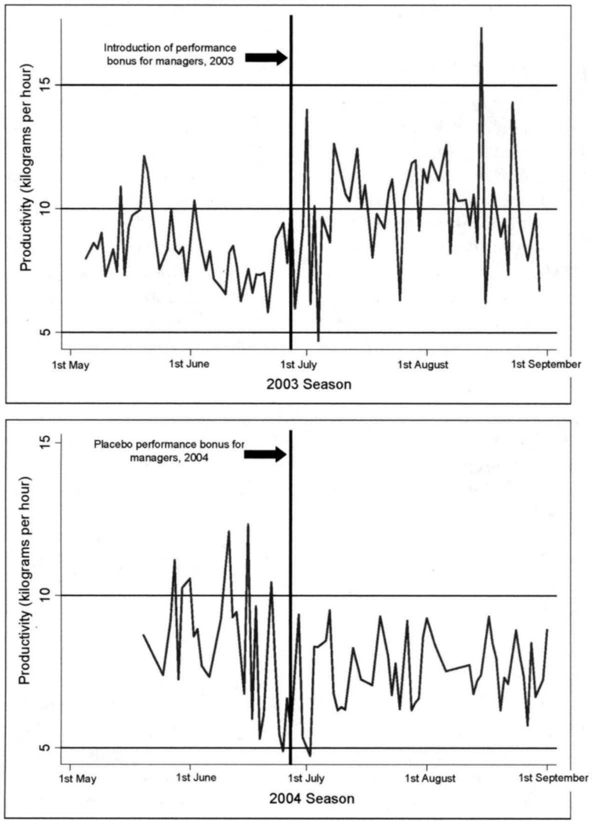

FIGURE I
Time Series of Productivity, 2003 and 2004 Seasons.
Notes: Since there might be more than one field operated per day, each field-day level observation is weighted by the number of pickers on the field-day to compute

(a) Figure I: Time Series of Productivity

TABLE I
DESCRIPTIVES OF WORKER PRODUCTIVITY BY MANAGERIAL INCENTIVE SCHEME

Managerial Incentive Scheme, 2003 Season	Fixed wages (May 1st–June 26th) (1) All workers	Performance bonus (June 27th–August 31st) (2) All workers	Fixed wages (May 1st–June 26th) (3) Selected workers	(4) Fired workers	Performance bonus (June 27th–August 31st) (5) Selected workers
Worker's productivity (kg/hr)					
Mean	8.37	10.4	8.52	7.69	10.4
Sd, overall	4.29	5.99	4.45	3.44	5.99
Sd, between	2.43	3.35	2.49	2.11	3.35
Sd, within	3.48	4.64	3.58	2.98	4.64
Number of workers	197	130	130	67	130
Managerial Incentive Scheme, 2004 Season	All workers	All workers			
Worker's productivity (kg/hr)					
Mean	7.86	7.85			
Sd, overall	5.24	3.51			
Sd, between	3.08	2.20			
Sd, within	4.21	2.87			
Number of workers	136	136			

Notes: These figures are based on all workers that are available for work three weeks either side of the change in managerial incentive schemes. Selected workers are define to be those that pick at least one field-day under both managerial incentive schemes. Fired workers are only selected to pick when managers are paid fixed wages. (All observatior are at the worker–field-day level.)

(b) Table I: Worker Productivity by Incentive Scheme

6.2 Figure II: Productivity Distribution and Scatter Plot + Table II: Descriptives by Incentive Scheme

Read: Figure II shows the density shifting right and becoming more dispersed. Table II shows productivity rising while kilograms per field-day remain about constant; hours fall from 3.70 to 3.03.

Economic message. The firm obtains faster picking rather than more fruit per worker-day; worker earnings remain roughly stable while piece rates fall.

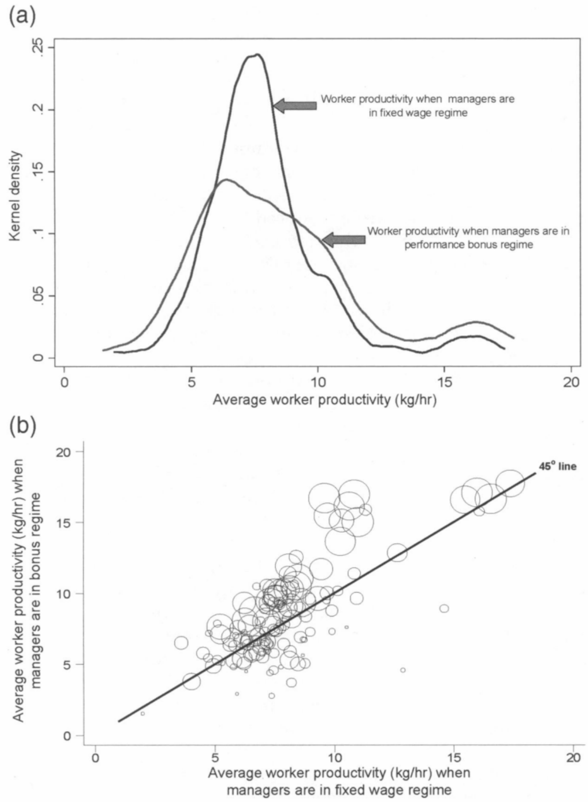

FIGURE II

(a) Kernel Density Estimates of Worker Productivity by Managerial Incentive Scheme (b) Scatter Plot of Worker Productivity by Managerial Incentive Scheme

Notes: Both figures use data on workers that are selected to pick fruit at least once under each managerial compensation schemes. The density estimates in Figure 2a are calculated using an Epanechnikov kernel. In Figure 2b, each observation is weighted by the number of field-days the worker picks under the managerial bonus scheme. A larger circle indicates that the worker picks on more

(a) Figure II: Density and Scatter

TABLE II

DESCRIPTIVES BY MANAGERIAL INCENTIVE SCHEME

	Managerial Incentive Scheme	
	Fixed wages	Performance bonus
Worker productivity (kg/hr)	8.37	10.4
	(0.240)	(0.486)
Kilograms picked per field-day	30.2	30.4
	(0.873)	(1.54)
Hours worked per field-day	3.70	3.03
	(0.169)	(0.157)
Hourly earnings from picking (£/hr)	4.81	4.53
	(0.133)	(0.199)
Piece rate per kilogram picked (£/kg)	0.617	0.476
	(0.030)	(0.016)
Number of workers on field-day	79.3	56.4
	(4.02)	(2.02)
Number of managers on field-day	5.27	3.28
	(0.231)	(0.075)
Worker-manager ratio	21.3	19.2
	(2.06)	(0.622)

Notes: Worker productivity, kilos picked per field-day, and hourly earnings are all calculated at the worker–field-day level. The standard errors on these worker–field-day level variables are clustered at the worker level. Hours worked per field-day, the piece rate per kilogram picked, the number of managers on the field-day, the number of workers on the field-day, and the worker–manager ratio, are all calculated at the field-day level.

Values given are means, standard errors in parentheses.

(b) Table II: Descriptives

6.3 Table III: Mean Productivity Regressions + Table IV: Dispersion Regressions

Read: Table III estimates a 0.19–0.23 log productivity effect. Table IV shows positive effects on the coefficient of variation.

Economic message. Managerial performance pay raises average productivity and inequality at the same time.

TABLE III

THE EFFECT OF THE MANAGERIAL INCENTIVES ON AVERAGE WORKER PRODUCTIVITY, FIELD-DAY LEVEL (Dependent Variable = Log of average productivity (kilogram picked per hour on field-day))

	(1) OLS	(2) Controls	(3) Field specific AR(1)	(4) Manager fixed effects	(5) Shorter window	(6) Tenure
Managerial performance bonus dummy	.225***	.203***	.196***	.194***	.195***	.190**
	(.044)	(.074)	(.069)	(.082)	(.095)	(.082)
Field life cycle		−1.35***	−1.42***	−1.31***	−1.38**	−1.29***
		(.167)	(.194)	(.177)	(.251)	(.174)
Average picking experience of workers		.284***	.276***	.313***	.352***	.335***
		(.050)	(.065)	(.062)	(.094)	(.093)
Time trend		−.003	−.002	−.001	−.002	−.003
		(.002)	(.002)	(.002)	(.007)	(.006)
Tenure under performance bonus scheme						.002
						(.005)
Field fixed effects	No	Yes	Yes	Yes	Yes	Yes
Manager fixed effects	No	No	No	Yes	Yes	Yes
R-squared	.0986	.3873	.8264	.8746	.8829	.8759
Number of field-day observations	247	247	247	247	171	247

Notes: *** denotes significance at 1 percent, ** at 5 percent, and * at 10 percent. All continuous variables are in logarithms. OLS regression estimates are reported in column (1) and (2). Robust standard errors are calculated. In the remaining columns AR(1) regression estimates are reported. Panel corrected standard errors are calculated using Prais-Winsten regression. This allows the error terms to be field specific heteroskedastic, and contemporaneously correlated across fields. The autocorrelation process is assumed to be specific to each field. Each field-day observation is weighted by the log of the number of workers present. The managerial performance bonus dummy = 1 when the manageria performance bonus scheme is in place, and 0 otherwise. The field life cycle is defined as the nth day the field is picked divided by the total number of days the field is picked ove the season. Tenure under the performance bonus scheme is defined as the number of field-days the performance bonus has been in place for. The specification in column (5) restrict the sample to days up to and including the first 9 days in the post bonus period. These are the dates over which the piece rate has not adjusted to the higher productivity of workers

(a) Table III: Average Productivity

TABLE IV

THE EFFECT OF THE MANAGERIAL INCENTIVES ON THE DISPERSION OF WORKERS' PRODUCTIVITY, FIELD-DAY LEVEL (Dependent Variable = Log of the coefficient of variation of productivity (kilogram picked per hour on field-day). Standard errors allow for field specific AR(1))

	(1) OLS	(2) Controls	(3) Field specific AR(1)	(4) Manager fixed effects	(5) Tenure
Managerial performance bonus dummy	.084***	.177***	.191***	.317***	.314***
	(.031)	(.060)	(.058)	(.063)	(.065)
Field life cycle		.024	.040	.208	.228
		(.150)	(.135)	(.137)	(.145)
CV of picking experience of workers		−.029	−.016	−.082	−.077
		(.081)	(.079)	(.072)	(.073)
Time trend		−.002	−.002	−.001	−.002
		(.001)	(.002)	(.002)	(.003)
Tenure under performance bonus scheme					.001
					(.003)
Field fixed effects	No	Yes	Yes	Yes	Yes
Manager fixed effects	No	No	No	Yes	Yes
R-squared	.0279	.0731	.5364	.5780	.5812
Number of field-day observations	247	247	247	247	247

Notes: *** denotes significance at 1 percent, ** at 5 percent, and * at 10 percent. All continuous variables are in logarithms. OLS regression estimates are reported in columns (1) and (2). Robust standard errors are calculated. In the remaining columns, AR(1) regression estimates are reported. Panel corrected standard errors are calculated using a Prais-Winsten regression. This allows the error terms to be field specific heteroskedastic and contemporaneously correlated across fields. The autocorrelation process is assumed to be specific to each field. Each field-day observation is weighted by the log of the number of workers present. The managerial performance bonus dummy = 1 when the managerial performance bonus scheme is in place, and 0 otherwise. The field life cycle is defined as the nth day the field is picked divided by the total number of days the field is picked over the season. Tenure under the performance bonus scheme is defined as the number of field-days the performance bonus has been in place for.

(b) Table IV: Dispersion

6.4 Figure III: Quantile Regression and Fixed Effects + Table V: Selection into the Workforce

Read: Figure III shows stronger productivity gains at higher quantiles and among higher-ability workers. Table V shows selected-in workers rising from 9.03 to 11.11 kg/hr, while selected-out workers do not improve and fired workers have the lowest fixed-wage productivity.

Economic message. High-ability workers both receive more managerial attention and get selected into work more often.

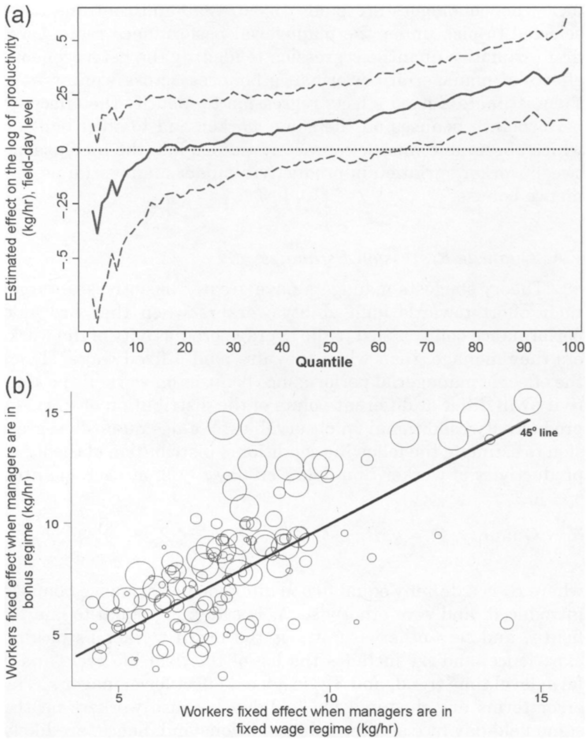

FIGURE III

(a) Quantile Regression Estimates (b) Workers' Fixed Effects

Notes: Figure 3a graphs the estimated effect of the managerial performance bonus dummy on the log of worker productivity at each quantile of the conditional distribution of the log of worker productivity and the associated 95 percent confidence interval. Bootstrapped standard errors that are clustered by field-day are estimated based on 1,000 replications. Figure 3b is based on a worker–field-day fixed effects regression. It plots the exponent of the workers fixed effect when managers are in the fixed wage regime against the exponent of their fixed effect when managers are in the performance bonus regime. Each observation is weighted by the number of field days the worker picks under the managerial

(a) Figure III: Quantile and Fixed Effects

TABLE V

SELECTION INTO THE WORKFORCE

	Selected-in workers	Selected-out workers	Fixed workers
A: Productivity[a]			
Fixed wages	9.03	7.45	6.79
	(3.03)	(2.09)	(2.15)
Performance bonus	11.11	7.35	
	(3.66)	(2.50)	
B: Unemployment rate[b]			
Fixed wages	.037	.089	.187
	(.052)	(.122)	(.186)
Performance bonus	.059	.146	.340
	(.060)	(.180)	(.372)

Notes: These figures are based on the sample of all 197 workers available to pick fruit. Selected-in workers are defined to be those that are in the top quartile of the distribution of number of field-days picked post-bonus. This corresponds to 77 or more field-day observations on which the worker picks post-bonus. Selected-out workers are defined to be those workers in the bottom three quartiles of the distribution of number of field-days picked post-bonus. Fired workers are those who never pick after the introduction of the performance bonus. There are 67 fired workers. The unemployment rate for a worker is the share of days in which the worker is present on the farm but is not assigned to any task.

a. Average productivity of workers (kg/hr) by worker type and managerial incentive scheme. Standard deviation in parentheses.

b. Average unemployment rate of workers by worker type and managerial incentive scheme. Standard deviation in parentheses.

(b) Table V: Selection into Workforce

6.5 Figure IV: Distribution of Field-days Selected + Figure V: Selection/Productivity and Unemployment/Selection

Read: Figure IV shows selection becoming more polarized under bonuses. Figure V shows that workers with larger productivity effects are selected more often and workers selected less often are more likely to be unemployed.

Economic message. Selection and targeting reinforce each other, translating productivity heterogeneity into earnings inequality.

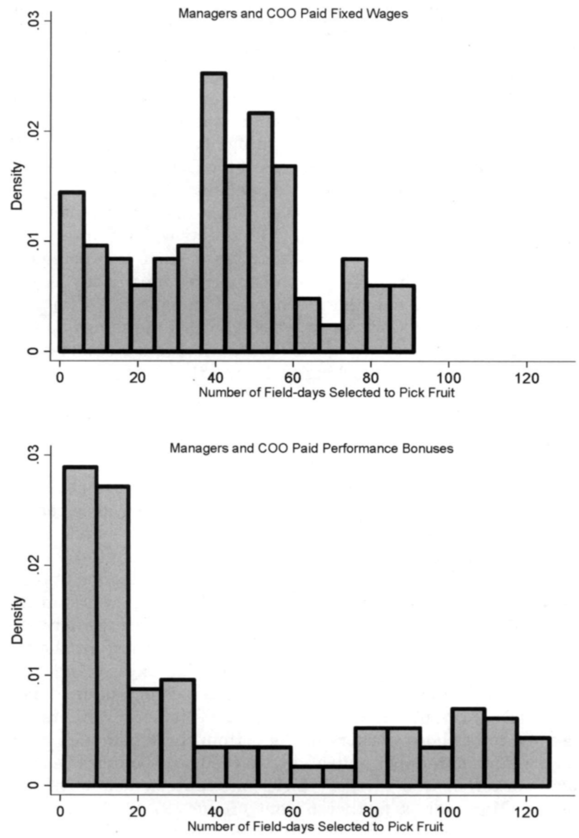

FIGURE IV
Distribution of Field-days Selected to Pick Fruit across Workers, by Managerial Incentive Scheme.
Notes: These histograms are drawn for those workers that are selected to pick fruit at least on one field-day under each managerial incentive scheme. Hence.

(a) Figure IV: Field-days Selected

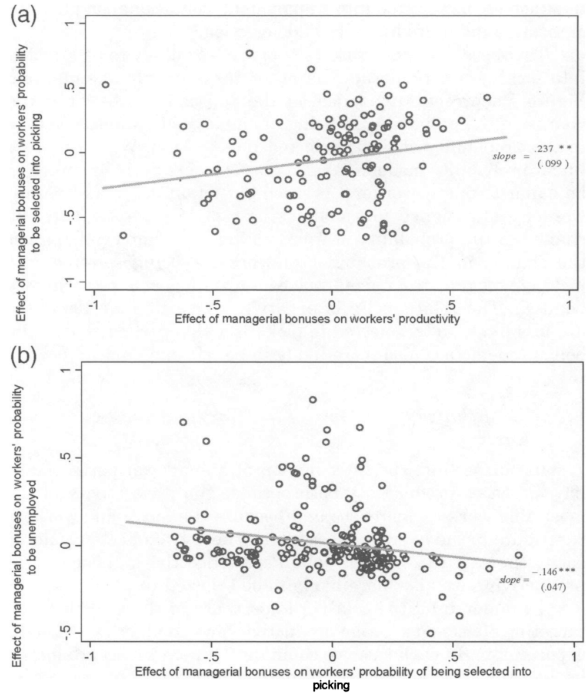

FIGURE V
(a) Selection and Productivity (b) Unemployment and Selection
Notes: To estimate the effect of the performance bonus on individual worker productivity, we regress log productivity on worker's picking experience, the field life cycle, a time trend and workers' fixed effects interacted with the bonus dummy. The effect of managerial bonuses on workers' productivity for any given worker is computed as the difference between the worker's fixed effect when managers are paid bonuses and the worker's fixed effect when managers are paid fixed wages. To estimate the effect of the performance bonus on the probability of being selected to pick fruit, we first define a selection dummy which is equal to one on days in which the worker is selected to pick, and zero otherwise. We then regress this selection dummy on labor supply, labor demand and workers' fixed effects interacted with the bonus dummy. The effect of managerial bonuses on

(b) Figure V: Selection and Unemployment

7 Short summary

- Managerial performance pay raises average productivity by about 21 percent.
- It also raises dispersion by increasing the productivity of high-ability workers more than others.
- At least half of the mean productivity gain comes from selection.
- The increase in dispersion is mostly due to targeting.
- The paper connects managerial incentive design to within-firm inequality.

8 Conclusion

8.1 Core conclusion

Managerial incentives affect both firm efficiency and worker inequality. Performance pay raises average productivity but reallocates managerial attention and work opportunities toward high-ability workers.

8.2 Economic intuition

When managers are paid for average productivity, their marginal return to helping the highest-productivity workers rises. The COO similarly has incentives to select more productive workers into picking. The same contract that improves performance also magnifies ability differences.

8.3 Incremental contribution revisited

The paper contributes by studying managerial rather than worker incentives, using a natural field experiment, and decomposing the effect into targeting and selection mechanisms.

8.4 Limitations and external validity

The setting is clean but specific: workers operate independently, output is objective, and the labor pool is fixed. In team production or open labor markets, effects may differ.

8.5 Takeaway

Incentive pay at one level of a firm can reshape productivity and earnings at another level. Incentive design is therefore also inequality design.